

Unimodality of inverse Z -polynomials of matroids

Rethlas

Abstract

We present a proof that the inverse Z -polynomial of every loopless matroid has unimodal coefficients, as conjectured by Gao, Ruan, and Xie. The proof rests on an identity relating the inverse Z -polynomial of a matroid to the inverse Kazhdan–Lusztig polynomials of the matroid and its truncation. More precisely, the successive differences of the coefficients up to the middle are sums of coefficients of these two inverse Kazhdan–Lusztig polynomials. Their nonnegativity, proved by Braden, Huh, Matherne, Proudfoot, and Wang, therefore implies unimodality.

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1 Introduction

The Kazhdan–Lusztig polynomial $P_M(x)$ of a matroid M , introduced by Elias, Proudfoot, and Wakefield [EPW16], is a polynomial invariant determined by the lattice of flats. Proudfoot, Xu, and Young [PXY18] introduced the related Z -polynomial

$$Z_M(x) = \sum_{F \in L(M)} x^{\text{rk } F} P_{M/F}(x),$$

where $L(M)$ denotes the lattice of flats and M/F denotes contraction. Proudfoot, Xu, and Young proved that $Z_M(x)$ is symmetric [PXY18]. Braden, Huh, Matherne, Proudfoot, and Wang [BHMPW26] proved nonnegativity for both polynomials and unimodality for $Z_M(x)$ using singular Hodge theory. These results motivate analogous questions for the inverse polynomials.

Gao and Xie [GX21] introduced the inverse Kazhdan–Lusztig polynomial $Q_M(x)$. The inverse Z -polynomial first appeared, with a different sign convention, in the work of Ferroni, Matherne, Stevens, and Vecchi [FMSV24, p. 25, footnote 9]. Gao, Ruan, and Xie [GRX26] subsequently developed its theory and adopted the normalization with nonnegative coefficients. Throughout this paper we use that normalization. For a loopless matroid M of rank d on a ground set E , it is

$$Y_M(x) = \sum_{F \in L(M)} (-x)^{d-\text{rk } F} \mu_{L(M)}(F, E) Q_{M|F}(x), \quad (1.1)$$

where $M|F$ is restriction to F and $\mu_{L(M)}$ is the Möbius function. The word “inverse” refers to inversion in the incidence algebra of the lattice of flats, with a rank-dependent sign. We recall the characterization of Q_M in Section 2.

A sequence $(a_0, \dots, a_d)$ is *unimodal* if it weakly increases up to some index and then weakly decreases. Gao, Ruan, and Xie [GRX26, Conjecture 1.4] conjectured that $Y_M(x)$ has unimodal coefficients for every loopless matroid, and proved this for sparse paving matroids. We give a proof for arbitrary loopless matroids. The earlier solution of Luis Ferroni and the subsequent independent solution of Elias Badalov are acknowledged below; this manuscript is an archival record of Rethlas’s proof and makes no claim of priority.

Theorem 1.1. *Let M be a loopless matroid of rank d , and write $Y_M(x) = \sum_{i=0}^d y_i x^i$. Then*

$$y_i = y_{d-i} \quad (0 \leq i \leq d), \quad y_0 \leq y_1 \leq \dots \leq y_{\lfloor d/2 \rfloor}.$$

In particular, the coefficient sequence of $Y_M(x)$ is unimodal.

The symmetry in this statement is already known [GRX26, Lemma 2.2]. We prove both properties from a truncation identity. For $d \geq 2$, let $T(M)$ denote the rank- $(d-1)$ truncation of M , whose independent sets are the independent sets of M of cardinality at most $d-1$.

Theorem 1.2. *Let M be a loopless matroid of rank $d \geq 2$, let $T = T(M)$, and put*

$$A_M(x) = Q_M(x) + xQ_T(x).$$

Then

$$(1-x)Y_M(x) = A_M(x) - x^{d+1}A_M(x^{-1}). \quad (1.2)$$

Consequently, if $y_i = [x^i]Y_M(x)$ and $q_i(N) = [x^i]Q_N(x)$, then

$$y_i - y_{i-1} = q_i(M) + q_{i-1}(T) \quad \left(0 \leq i \leq \left\lfloor \frac{d}{2} \right\rfloor\right), \quad (1.3)$$

with $y_{-1} = q_{-1}(T) = 0$.

Theorem 1.2 identifies every successive difference up to the middle with a sum of two inverse Kazhdan–Lusztig coefficients. The nonnegativity theorem of Braden, Huh, Matherne, Proudfoot, and Wang [BHMPW26, Theorem 1.5] then gives Theorem 1.1. The argument uses the inverse Kazhdan–Lusztig functional equation and the behavior of the characteristic polynomial under truncation. It applies without a representability assumption.

1.1 Related work

We review the development of the inverse polynomials and the results most directly related to the conjecture. Gao and Xie [GX21] characterized Q_M recursively, proved multiplicativity, computed the constant term and the polynomials of uniform matroids, and conjectured coefficientwise nonnegativity and log-concavity. Proudfoot [Pro21] developed the equivariant incidence-algebra framework and introduced equivariant inverse Kazhdan–Lusztig polynomials. Braden, Huh, Matherne, Proudfoot, and Wang [BHMPW26] proved nonnegativity for all matroids, including an equivariant version. Their theorem is the positivity input in our proof.

The inverse Z -polynomial arose in the work of Ferroni, Matherne, Stevens, and Vecchi [FMSV24] on Hilbert–Poincaré series of matroid Chow rings and intersection cohomology.

Gao, Ruan, and Xie [GRX26] established nonnegativity, palindromicity, multiplicativity, and valuativity for the normalized polynomial Y_M . Their valuativity argument builds on the theory of matroid valuations developed by Ardila and Sanchez [AS23] and Ferroni and Schröter [FS24]. They obtained formulas for uniform and sparse paving matroids and proved both unimodality and log-concavity in the sparse paving case [GRX26, Theorem 1.6]. They also formulated the general unimodality and log-concavity conjectures [GRX26, Conjectures 1.4 and 1.5]. Theorem 1.1 proves the unimodality statement; the coefficient difference formula does not by itself establish log-concavity.

Braden, Ferroni, Matherne, and Nepal [BFMN25] derived deletion formulas for both Q_M and Y_M , extending the deletion approach of Braden and Vysogorets [BV20] for the ordinary polynomials. They used these formulas to study projective geometries and matroids of corank two, providing further tools for coefficient questions. Gao, Li, and Xie [GLX26] introduced equivariant inverse Z -polynomials, proved that their coefficients are honest representations and that they are palindromic, and obtained formulas for uniform, q -niform, and paving matroids. They proved equivariant unimodality and strong induced log-concavity for uniform and q -niform matroids, and conjectured equivariant unimodality in general [GLX26, Conjecture 1.2]. Gao, Li, and Li [GLL26] computed generating functions and explicit formulas for the inverse Kazhdan–Lusztig and inverse Z -polynomials of fan matroids; they also proved log-concavity of the inverse Kazhdan–Lusztig coefficients for that family.

Related work on Q_M supplies formulas and further coefficient inequalities. Vecchi [Vec21] studied its relation to matroid modularity and its leading coefficient. Ferroni and Vecchi [FV22] studied circuit-hyperplane relaxation, and Ferroni, Nasr, and Vecchi [FNV23] used stressed-hyperplane relaxation to obtain formulas for paving matroids. Karn, Nasr, Proudfoot, and Vecchi [KNPV23] extended the relaxation formulas to the equivariant setting. Gao, Xie, and Yang [GXY22] computed equivariant inverse Kazhdan–Lusztig polynomials of uniform matroids. Gao, Li, Xie, Yang, and Zhang [GLXYZ26] established induced log-concavity results for equivariant invariants of uniform and q -niform matroids, and Xie and Zhang [XZ26] proved log-concavity of Q_M for paving matroids. Gao, Li, and Xie [GLX25] also obtained equivariant inverse Kazhdan–Lusztig formulas and log-concavity results for thagomizer matroids.

A further connection comes from the Chow functions for partially ordered sets developed by Ferroni, Matherne, and Vecchi [FMV24]. Caiolo, Ferroni, and Hoster [CFH26, Section 7.3] identify Q_M and Y_M with the left Kazhdan–Lusztig–Stanley polynomial and the Z -polynomial associated to a sign-twisted reverse characteristic kernel. This places the inverse polynomials in a broader framework of polynomial invariants of ranked posets.

Section 2 fixes conventions and recalls the inverse Kazhdan–Lusztig inputs. Section 3 proves Theorem 1.2. Section 4 deduces Theorem 1.1 and gives a nonnegative expansion of Y_M in symmetric unimodal polynomials.

1.2 Declaration of generative AI usage

The initial proof was generated autonomously by Rethlas, the AI agent for mathematical research [Ju+26], using GPT-5.6 Sol Max as its base model. AI assistance was also used to prepare and edit this manuscript. Ronnie Cheng and Shurui Liu verified the proof.

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Rethlas [Ju+26] was developed by Haocheng Ju, Jiedong Jiang, Shurui Liu, Guoxiong Gao, Yuefeng Wang, Zeming Sun, Leheng Chen, and Bin Wu, under the guidance of Professors Liang Xiao and Bin Dong.

We thank Luis Ferroni for suggesting the problem to Shurui Liu in a private email correspondence and Ferroni had independently solved the problem without AI assistance before Rethlas obtained its proof. Later, Elias Badalov obtained an independent solution with AI assistance. This manuscript makes no claim to priority or credit for resolving the conjecture; its purpose is to preserve Rethlas's proof to this conjecture as an archival record for the community.

2 Preliminaries

All matroids in this paper are finite and loopless. For a matroid N on E_N , we write $L(N)$ for its lattice of flats, ordered by inclusion, and rk_N for its rank function. When the ambient matroid is clear, we abbreviate $\text{rk}_N(F)$ to $\text{rk } F$. The minimum and maximum elements of $L(N)$ are $\emptyset$ and E_N . The Möbius function is characterized by

$$\mu(F, F) = 1, \quad \sum_{F \leq G \leq H} \mu(F, G) = 0 \quad (F < H).$$

For $e = \text{rk } N$, our convention for the characteristic polynomial is

$$\chi_N(x) = \sum_{F \in L(N)} \mu_{L(N)}(\emptyset, F) x^{e - \text{rk } F}. \quad (2.1)$$

We also write $\mu_N = \mu_{L(N)}(\emptyset, E_N)$. If F is a flat, then $N|F$ and N/F are loopless, and their lattices of flats identify with $[\emptyset, F]$ and $[F, E_N]$, respectively. In particular,

$$\mu_{N/F} = \mu_{L(N)}(F, E_N). \quad (2.2)$$

We use the following characterization of inverse Kazhdan–Lusztig polynomials.

Theorem 2.1 (Gao–Xie [GX21, Theorem 1.2]). *There is a unique assignment of a polynomial $Q_N(x) \in \mathbb{Z}[x]$ to each loopless matroid N such that:*

- (i) *if $\text{rk } N = 0$, then $Q_N(x) = 1$;*
- (ii) *if $\text{rk } N > 0$, then $\deg Q_N(x) < \text{rk } N/2$;*
- (iii) *if $e = \text{rk } N$, then*

$$(-x)^e Q_N(x^{-1}) = \sum_{F \in L(N)} (-1)^{\text{rk } F} Q_{N|F}(x) x^{e - \text{rk } F} \chi_{N/F}(x^{-1}). \quad (2.3)$$

Theorem 2.2 (Braden–Huh–Matherne–Proudfoot–Wang [BHMPW26, Theorem 1.5]). *For every loopless matroid N , the coefficients of $Q_N(x)$ are nonnegative integers.*

We will write $q_i(N) = [x^i]Q_N(x)$, taking $q_i(N) = 0$ when $i < 0$ or $i > \deg Q_N$. Multiplying (2.3) by $(-1)^e$ gives the equivalent form

$$x^e Q_N(x^{-1}) = \sum_{F \in L(N)} (-1)^{e - \text{rk } F} Q_{N|F}(x) x^{e - \text{rk } F} \chi_{N/F}(x^{-1}), \quad (2.4)$$

which will simplify the signs in the proof.

For completeness, the rank-zero matroid has $Y_N(x) = 1$. If N has rank one, its lattice of flats has two elements and $\chi_N(x) = x - 1$. The degree bound forces $Q_N(x) = c$ to be constant; (2.3) becomes

$$-xc = 1 - x - c,$$

so $c = 1$ and $Y_N(x) = 1 + x$.

3 A truncation identity

Let N be a loopless matroid of rank $s \geq 2$. The truncation $T(N)$ has the same ground set and rank function

$$\text{rk}_{T(N)}(X) = \min\{\text{rk}_N(X), s - 1\}. \quad (3.1)$$

Its proper flats are precisely the flats of N of rank at most $s - 2$. In particular, their lower intervals in the lattice of flats are unchanged. We first record the resulting identity for characteristic polynomials.

Lemma 3.1. *For a loopless matroid N of rank $s \geq 2$,*

$$x\chi_{T(N)}(x) = \chi_N(x) + (x - 1)\mu_N. \quad (3.2)$$

Proof. Write $\mu_N(G) = \mu_{L(N)}(\emptyset, G)$ and let $\mathcal{H}(N)$ denote the set of hyperplanes of N . Since lower intervals at the common proper flats are unchanged, the Möbius recursions at the top of the two lattices give

$$\mu_{T(N)} = - \sum_{\text{rk } G \leq s-2} \mu_N(G) = \mu_N + \sum_{H \in \mathcal{H}(N)} \mu_N(H).$$

In the difference $x\chi_{T(N)}(x) - \chi_N(x)$, all terms indexed by flats of rank at most $s - 2$ cancel. The remaining terms are

$$x\mu_{T(N)} - x \sum_{H \in \mathcal{H}(N)} \mu_N(H) - \mu_N = (x - 1)\mu_N,$$

as required. $\square$

Lemma 3.2. *Let M be a loopless matroid of rank $d \geq 2$, and put $T = T(M)$. For every flat F with $\text{rk } F \leq d - 2$,*

$$T|F = M|F, \quad T/F = T(M/F).$$

Proof. The rank function is unchanged on every subset of F , which proves the first equality. If $r = \text{rk } F$ and $X \subseteq E \setminus F$, then

$$\text{rk}_{T/F}(X) = \min\{\text{rk}_M(X \cup F), d - 1\} - r = \min\{\text{rk}_{M/F}(X), d - r - 1\}.$$

This is the rank function of $T(M/F)$. $\square$

We can now compare the inverse Kazhdan–Lusztig recursions of M and $T(M)$. The ranks differ by one, so adding the normalized recursions compares the two characteristic polynomials at each common proper flat.

Proof of Theorem 1.2. Set $T = T(M)$, and for $F \in L(M)$ write

$$r_F = \text{rk } F, \quad s_F = d - r_F, \quad \mu_F = \mu_{L(M)}(F, E), \quad Q_F(x) = Q_{M|F}(x).$$

If $s_F \geq 2$, Lemmas 3.1 and 3.2, applied to M/F , give

$$x^{s_F-1} \chi_{T/F}(x^{-1}) = x^{s_F} \chi_{M/F}(x^{-1}) + (x^{s_F-1} - x^{s_F}) \mu_F. \quad (3.3)$$

Apply (2.4) to M and to T , and add the resulting equations. The two top-flat terms are Q_M and Q_T . Each hyperplane H of M occurs only in the first sum; since M/H has rank one, its contribution is

$$-x Q_H(x) \chi_{M/H}(x^{-1}) = (x-1) Q_H(x).$$

For a common proper flat F , the signs in the two sums are $(-1)^{s_F}$ and $(-1)^{s_F-1}$. Equation (3.3) therefore shows that their combined contribution is

$$(-1)^{s_F} \mu_F (x^{s_F} - x^{s_F-1}) Q_F(x).$$

Multiplying the sum of the two recursions by x yields

$$\begin{aligned} x^{d+1} Q_M(x^{-1}) + x^d Q_T(x^{-1}) &= x(Q_M(x) + Q_T(x)) + x(x-1) \sum_{s_F=1} Q_F(x) \\ &\quad + (x-1) \sum_{s_F \geq 2} (-x)^{s_F} \mu_F Q_F(x). \end{aligned}$$

For a hyperplane F , one has $\mu_F = -1$, so $(-x)^{s_F} \mu_F = x$. Hence the last two sums combine to $(x-1)(Y_M - Q_M)$, by (1.1). We obtain

$$\begin{aligned} x^{d+1} Q_M(x^{-1}) + x^d Q_T(x^{-1}) &= x(Q_M + Q_T) + (x-1)(Y_M - Q_M) \\ &= Q_M + x Q_T - (1-x) Y_M. \end{aligned}$$

This is (1.2).

To extract coefficients, put $k = \lfloor d/2 \rfloor$. If $d = 2k$, the inverse Kazhdan–Lusztig degree bounds give

$$\deg Q_M \leq k-1, \quad \deg Q_T \leq k-1;$$

if $d = 2k+1$, they give

$$\deg Q_M \leq k, \quad \deg Q_T \leq k-1.$$

In both cases, $\deg A_M \leq k$, so every monomial of $x^{d+1} A_M(x^{-1})$ has degree at least $d+1-k > k$. Taking the coefficient of x^i in (1.2), for $0 \leq i \leq k$, gives

$$y_i - y_{i-1} = [x^i] A_M(x) = q_i(M) + q_{i-1}(T),$$

which proves (1.3). □

4 Unimodality and coefficient formulas

Proof of Theorem 1.1. The rank-zero and rank-one cases follow from $Y_M(x) = 1$ and $Y_M(x) = 1 + x$, respectively. Assume $d \geq 2$ and put $T = T(M)$.

First, the right-hand side of (1.2) changes sign under the transformation $f(x) \mapsto x^{d+1}f(x^{-1})$. Applying this transformation to both sides gives

$$-(1-x)x^d Y_M(x^{-1}) = -(1-x)Y_M(x).$$

Cancelling $1-x$ shows that

$$x^d Y_M(x^{-1}) = Y_M(x). \quad (4.1)$$

Thus $y_i = y_{d-i}$.

By (1.3) and Theorem 2.2,

$$y_i - y_{i-1} = q_i(M) + q_{i-1}(T) \geq 0 \quad \left(0 \leq i \leq \left\lfloor \frac{d}{2} \right\rfloor\right).$$

The coefficients therefore weakly increase up to the middle; (4.1) gives the weak decrease after the middle. $\square$

The truncation identity also gives an explicit nonnegative expansion in symmetric unimodal polynomials.

Corollary 4.1. *Let M be a loopless matroid of rank $d \geq 2$ and let $T = T(M)$. Then*

$$Y_M(x) = \sum_{i=0}^{\lfloor d/2 \rfloor} (q_i(M) + q_{i-1}(T)) x^i (1 + x + \cdots + x^{d-2i}). \quad (4.2)$$

In particular, for every $0 \leq j \leq d$,

$$y_j = \sum_{i=0}^{\min\{j, d-j\}} (q_i(M) + q_{i-1}(T)). \quad (4.3)$$

Proof. Write $A_M(x) = \sum_{i=0}^{\lfloor d/2 \rfloor} a_i x^i$, where $a_i = q_i(M) + q_{i-1}(T)$. Dividing (1.2) by $1-x$ gives

$$Y_M(x) = \sum_{i=0}^{\lfloor d/2 \rfloor} a_i \frac{x^i - x^{d+1-i}}{1-x} = \sum_{i=0}^{\lfloor d/2 \rfloor} a_i x^i (1 + x + \cdots + x^{d-2i}).$$

The summand indexed by i contributes to x^j exactly when $i \leq j \leq d-i$, which gives (4.3). $\square$

Each polynomial $x^i(1 + x + \cdots + x^{d-2i})$ is symmetric about degree $d/2$ and has a block of coefficients equal to one. Formula (4.2) thus makes unimodality transparent. It also describes equality between consecutive coefficients: for $1 \leq i \leq \lfloor d/2 \rfloor$,

$$y_i = y_{i-1} \iff q_i(M) = q_{i-1}(T) = 0.$$

Example 4.2. Suppose M has rank two and its simplification has m points. Then $\chi_M(x) = x^2 - mx + (m-1)$, and the inverse Kazhdan–Lusztig recursion gives $Q_M(x) = m-1$. The truncation has rank one, so $Q_T(x) = 1$. Consequently,

$$A_M(x) = (m-1) + x, \quad Y_M(x) = (m-1) + mx + (m-1)x^2.$$

Here the difference between the middle and either end coefficient is the constant coefficient of Q_T .

Remark 4.3. Theorem 1.2 is an identity for ordinary polynomial invariants. An equivariant analogue would require a corresponding comparison of equivariant characteristic polynomials under truncation. The equivariant unimodality conjecture of Gao, Li, and Xie [GLX26, Conjecture 1.2] is a natural further question.

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